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IGHV3-53 immunoglobulin heavy variable 3-53 [Homo sapiens (human)]

Gene ID: 28420, updated on 19-May-2026

Download Datasets

Summary

Official Symbol

IGHV3-53 provided by [HGNC](#)

Official Full Name

immunoglobulin heavy variable 3-53 provided by [HGNC](#)

Primary source

[HGNC:HGNC:5610](#)

See related

[Ensembl:ENSG00000211967](#) [IMGT/GENE-DB:IGHV3-53](#); [AllianceGenome:HGNC:5610](#)

Gene type

other

RefSeq status

VALIDATED

Organism

[Homo sapiens](#)

Lineage

Eukaryota; Metazoa; Chordata; Craniata; Vertebrata; Euteleostomi; Mammalia; Eutheria; Euarchontoglires; Primates; Haplorrhini; Catarrhini; Hominidae; Homo

Also known as

VH; IGHV353

Summary

Predicted to enable antigen binding activity. Predicted to be involved in immunoglobulin mediated immune response. Predicted to be located in extracellular region and plasma membrane. Predicted to be part of immunoglobulin complex. [provided by Alliance of Genome Resources, Jul 2025]

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[Transcripts and proteins table](#)

Genomic context

See IGHV3-53 in [Genome Data Viewer](#)

Location: 14q32.33

Annotation release	Status	Assembly	Chr	Location
RS_2025_08	current	GRCh38.p14 (GCF_000001405.40)	14	NC_000014.9 (106592676..106593126, complement)
RS_2025_08	current	T2T-CHM13v2.0 (GCF_009914755.1)	14	NC_060938.1 (100913186..100913636, complement)

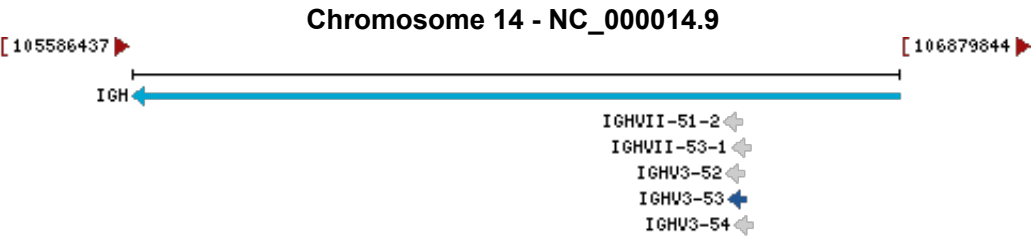
RS_2024_09

previous assembly

GRCh37.p13
([GCF_000001405.25](#))

14

NC_000014.8 (107048670..107049120, complement)

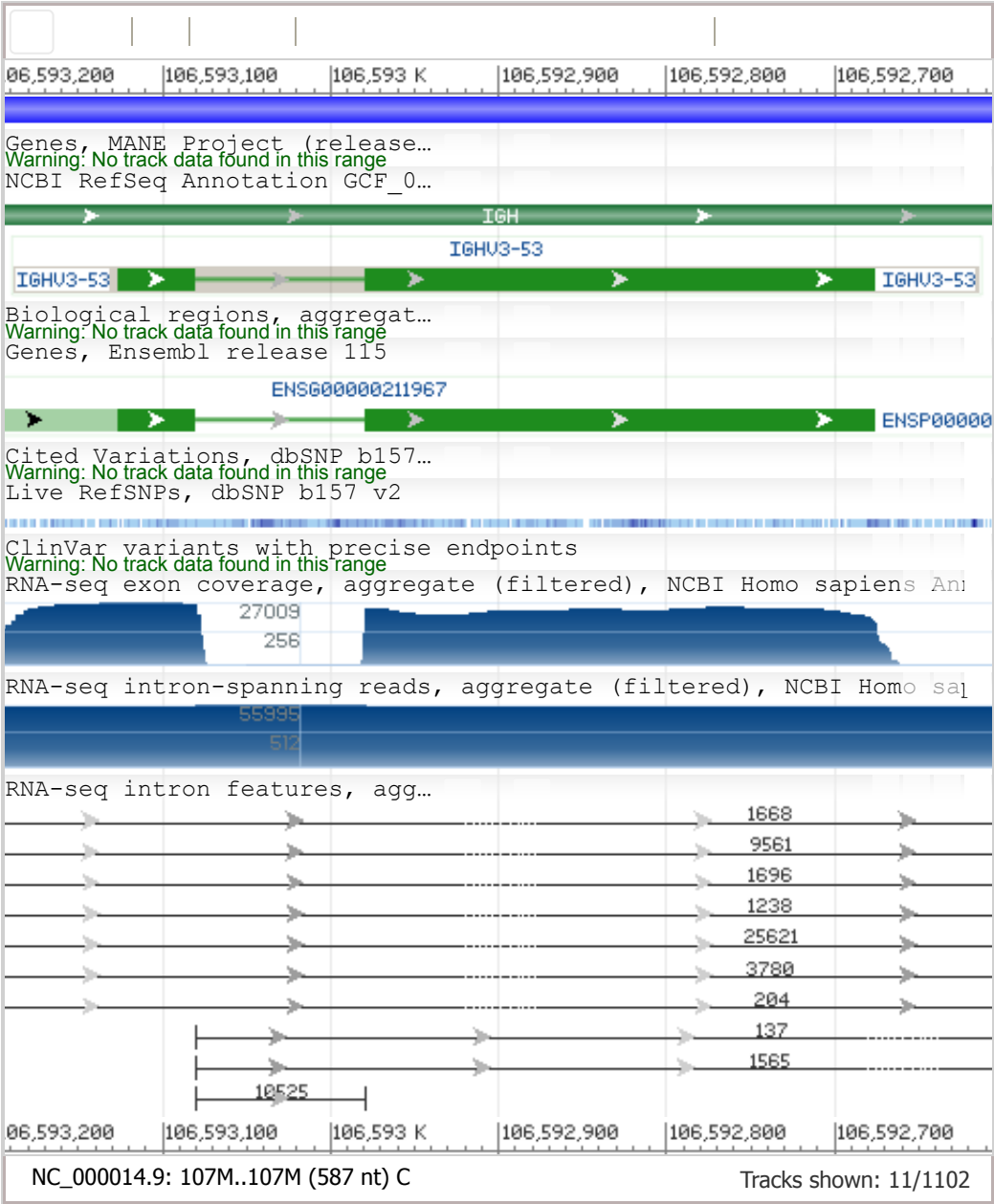


Genomic regions, transcripts, and products

Go to [reference sequence details](#)

Genomic Sequence: NC_000014.9 Chromosome 14 Reference GRCh38.p14 Primary Assembly

Go to nucleotide: [Graphics](#) [FASTA](#) [GenBank](#)



Bibliography

Related articles in PubMed

1. [Complete amino acid sequence of the alpha 2 heavy chain of a human IgA2 immunoglobulin of the A2m \(2\) allotype.](#)
Toraño A, *et al.* Proc Natl Acad Sci U S A, 1978 Feb. PMID 416441, [Free PMC Article](#)
2. [Paired heavy- and light-chain signatures contribute to potent SARS-CoV-2 neutralization in public antibody responses.](#)
Banach BB, *et al.* Cell Rep, 2021 Oct 5. PMID 34587480, [Free PMC Article](#)
3. [Structure and physical map of 64 variable segments in the 3'0.8-megabase region of the human immunoglobulin heavy-chain locus.](#)
Matsuda F, *et al.* Nat Genet, 1993 Jan. PMID 8490662
4. [The complete nucleotide sequence of the human immunoglobulin heavy chain variable region locus.](#)
Matsuda F, *et al.* J Exp Med, 1998 Dec 7. PMID 9841928, [Free PMC Article](#)
5. [Phylogenetic-based propagation of functional annotations within the Gene Ontology consortium.](#)
Gaudet P, *et al.* Brief Bioinform, 2011 Sep. PMID 21873635, [Free PMC Article](#)

GeneRIFs: Gene References Into Functions

[What's a GeneRIF?](#)

Submit: [New GeneRIF](#) [Correction](#)

Phenotypes

[BioGRID CRISPR Screen Phenotypes \(0 hits/29 screens\)](#)

[Review eQTL and phenotype association data in this region using PheGenI](#)

Variation

[See variants in ClinVar](#)

[See studies and variants in dbVar](#)

[See Variation Viewer \(GRCh37.p13\)](#)

Interactions

Products	Interactant	Other Gene	Complex	Source	Pubs	Description
BioGRID:118300	BioGRID:4383848	S	-	BioGRID	PubMed	Co-crystal Structure; Reconstituted Complex

General gene information

Related region gene

[Review record\(s\) in Gene](#)

Other Names

Ig heavy chain V-III region BUT
immunogloblin heavy chain variable region

Gene Ontology [Provided by GOA](#)

Function	Evidence Code	Pubs
enables antigen binding	IBA	
enables antigen binding	NAS	PubMed

Process	Evidence Code	Pubs
involved_ in immune response	NAS	PubMed
involved_ in immunoglobulin mediated immune response	IBA	

Component	Evidence Code	Pubs
located_ in extracellular region	IEA	
located_ in extracellular region	NAS	PubMed
located_ in extracellular region	TAS	
located_ in plasma membrane	IEA	
located_ in plasma membrane	TAS	

NCBI Reference Sequences (RefSeq)

NEW Try the new [Transcript table](#)

RefSeqs maintained independently of Annotated Genomes

These reference sequences exist independently of genome builds. [Explain](#)

Genomic

1. NG_001019.6

Range	287219..287669
Download	GenBank , FASTA , Sequence Viewer (Graphics)

RefSeqs of Annotated Genomes: GCF_000001405.40-RS_2025_08

The following sections contain reference sequences that belong to a specific genome build. [Explain](#)

Reference GRCh38.p14 Primary Assembly

Genomic

1. NC_000014.9 Reference GRCh38.p14 Primary Assembly

Range	106592676..106593126 complement
Download	GenBank , FASTA , Sequence Viewer (Graphics)

Reference GRCh38.p14 ALT_REF_LOCI_1

Genomic

1. NT_187600.1 Reference GRCh38.p14 ALT_REF_LOCI_1

Range	1084355..1084805 complement
Download	GenBank , FASTA , Sequence Viewer (Graphics)

Alternate T2T-CHM13v2.0

Genomic

1. NC_060938.1 Alternate T2T-CHM13v2.0

Range	100913186..100913636 complement
Download	GenBank , FASTA , Sequence Viewer (Graphics)

Related sequences

Nucleotide		Protein
Heading	Accession and Version	
genomic	AB019438.1	BAA75029.1
genomic	AC244452.3 (196879..197330)	None
genomic	CP068264.2 (100913186..100913636)	None
genomic	M99679.1	None

Protein Accession	Links	
	GenPept Link	UniProtKB Link
P01767.2	GenPept	UniProtKB/Swiss-Prot:P01767

Additional links

[Gene LinkOut](#)

